CellTrackAnalysis Instructions

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This is the **second step** of chemotaxis analysis in I2Tumbles. `CellTrackAnalysis.m` reads in a `.mat` file (the output from `SimpleTracking.m`) and provides chemotaxis metrics such as tumbles per second for the experimental group and individual bacterium as well. From the tracked X and Y coordinates from `SimpleTracking.m`, linear and angular velocity of each bacterium is derived. Then, the algorithm finds the local minima of the linear velocity and local maxima of angular velocity and find the tumble points based on the calculation (Johnson et al., 2024). It is important to understand how to interpret the results and adjust the parameters while using this algorithm.

Step 1: CellTrackAnalysis Parameters

1. Open `CellTrackAnalysis.m`
2. Adjust parameters. The most important parameters are listed below:

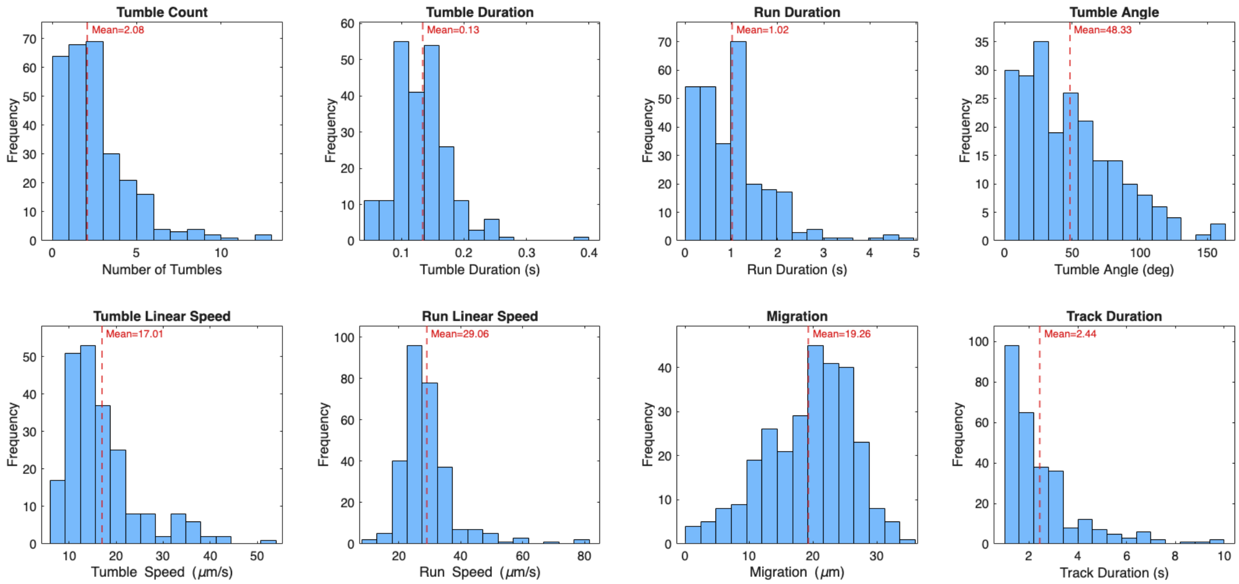
Parameter Name	Value Range	Data Type	Description
`secondVideo`	> 0	Integer	Duration of the video in second.
`depth_speed_ratio`	0.5 – 0.9999	Float	The ratio of linear speed vs local minima of the speed. The larger value will return less tumbles (more conservative). For more details, refer to Johnson et al. (2024)
`show_bacteria_labels`	true / false	Boolean	Setting this true will display bacteria ID numbers on the scatterplot output.
`fps_scale_sg_window`	true / false	Boolean	Setting this true will auto scale `sg_window` based on the video's FPS value. If set to false, will use the `sg_window` value input.
`sg_window`	Odd number	Integer	Window size in the Savitzky-Golay filter. A larger value will return less tumbles (more conservative). Highly recommend trying the `fps_scale_sq_window = true` then think about adjusting from auto-scaled number afterwards. For more details, Refer to sgolay function description in MATLAB.
`convert_to_um`	true / false	Boolean	Setting this to true will convert pixels to um.
`px_to_um`	> 0	Float	Conversion value: um per pixel
`min_tumble_speed`	> 0	Float	Minimum speed at tumble dip

Recommendations for adjusting the parameters:

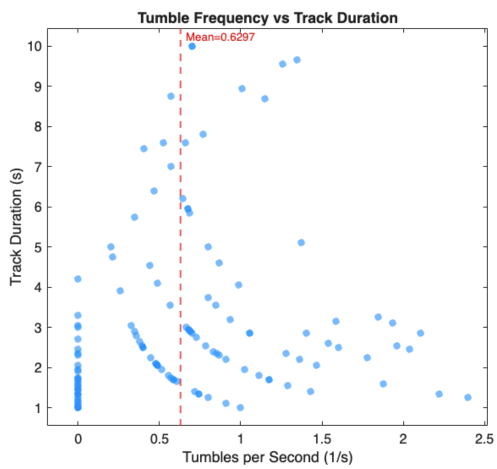
i. Set `secondVideo` correctly. Metrics like tumble per second will dramatically change if this value is not correct.
ii. Run the script with default value and run `PlotTrajectory.m` with `bacteria_idx` = 0. If the values do not make sense, adjust `depth_speed_ratio`.
iii. If the algorithm is still displaying too many tumbles, try increasing `sg_window` with `fps_scale_sg_window=false`. Only odd numbers can be used for `sg_window`.

3. Visualizations

   a. Histogram of metrics such as tumble count, tumble and run duration, tumble angle, linear speed, migration and track duration



   b. Scatterplot of tumble frequency vs track duration



4. Outputs

   a. Printout of summary metric of the experimental group

Population Summary (N = 131 bacteria analysed)

Metric	Mean	Median
Number of tumbles	2.11	1.00
Tumbles per second (1/s)	0.63	0.57
Tumble duration (s)	0.13	0.12
Run duration (s)	1.27	1.05
Tumble angle (deg)	44.86	38.62
Tumble linear speed (μm/s)	12.88	12.39
Run linear speed (μm/s)	22.98	22.25
Track duration (s)	3.02	2.35

b. CellTrackAnalysis_results.xlsx

i. An excel file with metrics per bacterium and summarized metrics

5. Full list of variables

Parameter Name	Value Range	Data Type	Description
`secondVideo`	> 0	Integer	Duration of the video in second.
`convert_to_um`	true / false	Boolean	Setting this to true will convert pixels to um.
`px_to_um`	> 0	Float	Conversion value: um per pixel
`fps_scale_sg_window`	true / false	Boolean	Setting this true will auto scale `sg_window` based on the video's FPS value. If set to false, will use the `sg_window` value input.
`sg_window`	Odd number	Integer	Window size in the Savitzky-Golay filter. A larger value will return less tumbles (more conservative). Highly recommend trying the `fps_scale_sq_window = true` then think about adjusting from auto-scaled number afterwards. For more details, Refer to sgolay function description in MATLAB.
`sg_poly`	2	Integer	Polynomial order of the function (keep at 2)
`minTrackLength`	sg_window + 2	Integer	Minimum track length to include in the calculation
`w_offset`	> 0	Integer	Rad/s offset added to the filter for overlay plot
`depth_speed_ratio`	0.5 – 0.9999	Float	The ratio of linear speed vs local minima of the speed. The larger value will return less tumbles (more conservative). For more details, refer to Johnson et al. (2024)
`v_min_threshold`	> 0	Float	Tumble frame width threshold Johnson et al. (2024)
`use_angular_confirmation`	true/ false	Boolean	Requires angular speed at the tumble frame to exceed the track-mean angular speed
`w_confirm_factor`	> 0	Float	The confirm factor number for angular speed at the tumble frame. Default value is 0.2. Johnson et al. (2024)
`min_tumble_speed`	> 0	Float	Minimum speed at tumble dip
`show_bacteria_labels`	true / false	Boolean	Setting this true will display bacteria ID numbers on the scatterplot output.

References

- Johnson et al. Run-and-tumble kinematics of Enterobacter Sp. SM3. *Physical Review.* E vol. 109,6–1 (2024). [link]
- Cell Track Analysis Algorithm GitHub: https://github.com/Tang-BioPhys/SM3-Motility/tree/main